

SKILL- Week 2

1.) Assignment-1

```
harshitha18@LAPTOP-6E22AR  X  +  v
harshitha18@LAPTOP-6E22AR6C:~$ mkdir OSSP
mkdir: OSSP: File exists
harshitha18@LAPTOP-6E22AR6C:~$ cd OSSP
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ git init
Reinitialized existing Git repository in /home/harshitha18/OSSP/.git/
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ mkdir src include obj bin docs tests scripts assets
mkdir: src: File exists
mkdir: include: File exists
mkdir: obj: File exists
mkdir: bin: File exists
mkdir: docs: File exists
mkdir: tests: File exists
mkdir: scripts: File exists
mkdir: assets: File exists
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch .gitignore README.md LICENSE Makefile
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch include/shell.h include/parser.h include/executor.h include/builtin.h
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch bin/my_shell
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch tests/test_parser.c tests/test_executor.c
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch scripts/run.sh
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ touch assets/architecture.png
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ tree
```

```
harshitha18@LAPTOP-6E22AR  X  +  v
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ tree
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   ├── my_shell
│   └── process_demo
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
│   ├── builtin.o
│   ├── executor.o
│   ├── main.o
│   ├── parser.o
│   └── shell.o
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   ├── process_demo
│   ├── process_demo.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

9 directories, 27 files
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ |
```

This assignment creates an organized OSSP project structure containing source files, header files, tests, documentation and scripts. `mkdir` is used to create directories, `touch` creates files, and `tree` displays the complete project structure in tree format.

2.) Assignment-2

Task-1;

```
harshitha18@LAPTOP-6E22AR  X + v
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ echo $$
224
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ ps
  PID TTY          TIME CMD
  224 pts/0        00:00:00 bash
  650 pts/0        00:00:00 ps
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ sleep 100 &
[1] 661
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ ps
  PID TTY          TIME CMD
  224 pts/0        00:00:00 bash
  661 pts/0        00:00:00 sleep
  671 pts/0        00:00:00 ps
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ pstree -p $$
bash(224)─pstree(681)
           └sleep(661)
```

Task-2;

```
harshitha18@LAPTOP-6E22AR  X + v
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ nano process.c
[1]+  Done                  sleep 100
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ gcc process.c -o process
harshitha18@LAPTOP-6E22AR6C:~/OSSP$ ./process
Enter command: ls
This is the Parent Process.
Parent PID = 784
Child PID = 785
This is the Child Process.
Child PID = 785
LICENSE  README.md  bin      include  process  scripts  tests
Makefile assets    docs    obj      process.c src
Child Process Completed.
```

`fork()` creates a child process from the parent process and gives both processes separate process IDs. `exec()` allows the child process to execute the command entered by the user while the parent waits for the child to complete.